Divyesh Vaghasiya

Ph.D. Candidate in Mathematics

Department of Mathematics and Statistics, University of South Florida

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Education

Aug. 2022 – present **Ph.D. in Mathematics (candidate)**, University of South Florida, Tampa, FL, USA. Expected completion: May 2027. Advisor: Dr. Lukas Kölsch. GPA: 4.00/4.00. Qualifying examinations passed: Real Analysis (May 2023), Algebra (January 2024), and Specialty (April 2025). Advanced to candidacy in 2025.

July 2020 – May 2022 **M.Sc. in Mathematics**, Indian Institute of Technology Bombay, Mumbai, India. GPA: 7.68/10.00. Reading project in analytic number theory, supervised by Prof. Ravi Raghunathan.

July 2017 – May 2020 **B.Sc. in Mathematics**, Ramnarain Ruia Autonomous College, Mumbai, India. GPA: 9.63/10.00.

Research Interests

Cryptographic functions over finite fields, including Almost Perfect Non-linear (APN), Perfect Non-linear (PN), and Bent functions; Additive Combinatorics and Finite Geometry; Algebraic Coding theory, including Rank-Metric (RM) and Locally Recoverable Codes (LRC); Secure Distributed Computation.

Developing interests: Quantum Coding theory and Quantum Distributed Computation.

Publications and Preprints

Peer-reviewed journal articles

[1] Sophie Hannah Bénéteau, Nicolas Goluboff, Lukas Kölsch, and **Divyesh Vaghasiya**. "On the Walsh Spectra of Quadratic APN Functions." *IEEE Transactions on Information Theory* **72**(7), 5207–5216, 2026. doi:10.1109/TIT.2026.3695003.

[2] Mohamed O. Darwish, Anuvrat Jaindungarwal, Giacomo Micheli, **Divyesh Vaghasiya**, and Shujun Zhao. "Good polynomials for locally recoverable codes: Classification results, asymptotics, and exact formulae." *Journal of Algebra and Its Applications* **24**(13–14), article 2541017, 2025. doi:10.1142/S0219498825410178.

Preprints

[3] Rafael G. L. D'Oliveira, Giulia Gaggero, Arturo Jaramillo Gil, Hiram H. López, Cecilia Martínez-Reyes, and **Divyesh Vaghasiya**. "Function Tables for Secure Distributed Matrix Multiplication." Preprint, 2026. arXiv:2609.08154.

Selected Research Projects

Ongoing **APN & PN functions, Sidon sets, and sum-free sets.**
With Dr. Lukas Kölsch.

Ongoing **"Good" Rational functions for Locally Recoverable Codes.**
With Anuvrat Jaindungarwal. Manuscript draft completed.

2026–present **Homogeneity Degree of Bent functions.**
With Dr. Lukas Kölsch and a team including undergraduate researchers.
Manuscript in final reading by the authors.

2024–2025 **Autotopisms of Finite Semifields and Rank-Metric Codes.**
With Dr. Lukas Kölsch.

Research Talks

Upcoming

Jan. 12, 2027 *Sidon and sum-free subgroups associated with APN and Planar exponents.*
Joint Mathematics Meetings, Chicago, IL, USA. Invited special-session presentation, scheduled.
AMS Special Session on Cryptography and Related Fields, I.

Delivered

Mar. 29, 2026 *Blocking Sets and Vector Space Partitions Related to Quadratic Functions over* $\mathbb{F}_2^n$.
AMS Spring Southeastern Sectional Meeting, Savannah, GA, USA. Special Session on Algebraic Combinatorics.

Mar. 29, 2026 *Good Polynomials and Rational Functions for Locally Recoverable Codes.*
AMS Spring Southeastern Sectional Meeting, Savannah, GA, USA. Special Session on Coding Theory and its Applications to Quantum Error-Correction and Cryptography.

Jan. 14, 2026 *On the Walsh Spectra of Quadratic APN Functions.*
Indo-European Conference on Mathematics, Pune, India. Contributed talk.

Research Funding

2026–2027 **Graduate Student Research Collaboration Award**, University of South Florida.
Project: *Analysis and design of code-based cryptosystems used in post-quantum cryptography*.
Principal investigator: Dr. Lukas Kölsch. **Role:** Graduate student researcher.
Total project award: $10,000; one year beginning July 1, 2026.

Scholarships and Awards

2026 **Best Team Presentation Award**, Gene Golub SIAM Summer School, Duke University.
Team: *Ready Player One*. **Topic:** Quantum Linear System Solvers.

2026 **M.V. Johns Jr. Scholarship for Graduate Study in Mathematics/Statistics**, University of South Florida. $500.
Graduate scholarship awarded on the basis of merit and financial need.

2026 **Graduate Student Travel Grant**, Department of Mathematics and Statistics, University of South Florida. $400 to support a Conference Presentation.

2024–2026 **Fred L. and Helen M. Tharp Endowed Scholarship**, University of South Florida.
Recognizes Teaching Assistants demonstrating outstanding progress toward their degree.
Awards: $1,382.43 (2026); $1,500 (2025); $1,300 (2024).

2020 **All India Rank 12 in Mathematics**, IIT Joint Admission Test for Masters (IIT JAM).

2017–2022 **INSPIRE Scholarship**, Department of Science and Technology, Government of India.
Five-year scholarship for higher education in science; awarded on the basis of performance within the top 1% of the Maharashtra State Board Class XII cohort.
INR 60,000 annually for five years (INR 300,000 total).

Teaching Experience

Aug. 2022–present **University of South Florida**, Tampa, FL, USA.
Graduate Teaching Associate, August 2024–present.
Graduate Instructional Assistant, August 2022–July 2024.

Primary instructor

Fall 2026 **Calculus I** (MAC 2311).

Teaching Assistant assignments

Elementary Abstract Algebra I (MAS 4301): Spring 2026.
Calculus II (MAC 2312): Spring 2026, Spring 2025, Fall 2023, Summer 2023.
Precalculus Algebra/Trigonometry (MAC 1147): Summer 2025.
Life Sciences Calculus I (MAC 2241): Summer 2025.
Calculus III (MAC 2313): Fall 2025, Fall 2024, Spring 2023.
Introduction to Discrete Structures (COT 3100): Spring 2025.
Differential Equations (MAP 2302): Fall 2024.
Computational Linear Algebra (MAS 3114): Spring 2024.
Engineering Calculus II (MAC 2282): Spring 2024, Spring 2023, Fall 2022.
Linear Algebra (MAS 3105): Fall 2023.
Bridge to Abstract Mathematics (MGF 3301): Summer 2023.
Calculus I (MAC 2311): Fall 2022.

Earlier teaching

May 2018–Sept. 2019 **Teacher**, Balaji Science & Engineering Private Tutorials, Mumbai, India.

May 2017–Mar. 2018 **Tutor**, Suresh Dani's Classes, Mumbai, India.

Research Mentoring

Summer 2026 **Mentor, RTG-supported Research Experience for Undergraduates**, University of South Florida.
Topic: Homogeneity Degree of Bent functions.
Project supervisor: Dr. Lukas Kölsch.
Undergraduate participants: Megan du Preez and Kovid Tandon.

Summer 2025 **Mentor, RTG-supported Research Experience for Undergraduates**, University of South Florida.
Topic: Quadratic APN functions and their Walsh spectra.
Project supervisor: Dr. Lukas Kölsch.
Undergraduate participants: Sophie Hannah Bénéteau and Nicolas Goluboff.

Workshops and Additional Training

July 27–Aug. 7, 2026 **Gene Golub SIAM Summer School: Fault-tolerant Algorithms in Quantum Computing**, Duke University, Durham, NC, USA.
Two-week program on mathematical foundations and advanced techniques in Quantum Computing.
Group project: Quantum Linear System Solvers.

June 15–19, 2026 **Workshop on Applications of Commutative Algebra and Coding Theory**, CIMAT, Guanajuato, Mexico.
Participant in the Secure Distributed Matrix Multiplication (SDMM) research working group.

Aug 2023–May 2024 **Guided study in Quantum Information theory**, University of South Florida.
Under the guidance of Prof. Kaiqi Xiong. Study and presentation of research literature in Quantum Linear Algebra and Quantum algorithms.

Service and Leadership

2019 **Event Head, MATHTRIX'19**, Ramnarain Ruia Autonomous College, Mumbai, India.
Led the annual mathematics festival; coordinated junior mentorship, faculty–student communication, sponsorships, and participation.

2017–2018 **Management Team Member, MATHTRIX**, Ramnarain Ruia Autonomous College, Mumbai, India.
Assisted with organizing the departmental mathematics festival in 2017 and 2018.

Skills and Languages

Programming and mathematical software: Python, SageMath, Magma, MATLAB.

Typesetting: LaTeX.

Languages: English, Hindi, Gujarati, and Marathi; reading, writing, and speaking.